Specifications

Higher · Design · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ explain the role of a specification in the design process
- ☐ write specific specification points from a brief and research
- ☐ describe the purpose and contents of a product design specification
- ☐ describe the purpose and contents of a performance specification
- ☐ describe the purpose and contents of a technical specification

I DO — Worked example

Read how a full-mark answer is built

1 Describe the purpose of a product design specification and a technical specification, giving an example of the information found in each. (4 marks)

Full-mark answer each ✓ is one mark

A product design specification sets out the parameters the design must meet and is used to evaluate decisions during development ✓, for example the target market and maximum retail price ✓. A technical specification is written after the design is finalised and gives the detail needed to manufacture it ✓, for example the materials, tolerances and assembly methods ✓.

Why marks are not awarded

"A PDS is a list of what the product needs. A technical spec is the technical details."

1 mark: the purposes are vague and no examples are given.

WE DO — Guided practice

Use the starters and word bank to help you

2 Describe the purpose of a performance specification and give two examples of what it contains. (3 marks)

Sentence starters

One … is …, which means …
Another … is …, so …
Also, … because …

Word bank

saying · designed · requirements · without · standards · perform · legislation · states

Tip: 3 marks = 3 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which is the strongest specification point?
☐ It must be safe for everyone to use ☐ It should look attractive to the target market ☐ It must be durable enough to last a long time ☐ It must fit a 40 mm diameter water bottle

b) "Must comply with BS EN 71 toy safety standards" belongs mainly in which specification?
☐ Technical ☐ Brief ☐ Performance ☐ Cutting list

c) Which specification is produced AFTER the product has been designed and tested?
☐ Product design ☐ Technical ☐ Performance ☐ Design brief

d) Which is most likely in a technical specification?
☐ Target market profile of 13–16 year olds ☐ It must appeal strongly to teenagers ☐ A retail price of under £20 in the shops ☐ Wall thickness of 2.5 mm in ABS

4 Key terms: write the letter of the matching description next to each term.

1. **Where does specification information come from?**	____	**A** The design brief and the results of research.
2. **Why is a specification important to a designer?**	____	**B** It sets clear boundaries, gives direction, and is used to evaluate ideas against the client and user needs.
3. **Performance specification**	____	**C** States the performance the product must meet without saying how: standards, legislation, sustainability, guarantees.

5 Explain what is meant by ‘Product design specification’. (2 marks)

6 Explain what is meant by ‘Technical specification’. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?